EE204 Semiconductor Devices

HW 4

1.

A p+/n silicon step junction is doped with N_A=10^{19}cm^{-3} and N_D=5x10^{15}cm^{-3}.

a) What is the built-in potential at room temperature?

b) What is the built-in potential at 800K?

c) What is the width of the depletion region at the n-side?

d) What is the total depletion region width of the P-N junction?

e) What is the maximum electric field in the depletion region?

f) If a reverse bias of 4V is applied to the diode, what is the new maximum electric field in the junction?

g) Calculate the percentage increase in the depletion width on the n-side and the p-side under 4V reverse bias.

(Use n_i = 10^{10} cm^{-3} at 300K)

[0.877 V, 0.209 V, 47.26 μm, 3.614 × 10^4 V/cm, 8.627 × 10^4 V/cm, 139%, 139%]

2.

An abrupt silicon pn junction at zero bias has dopant concentrations of $N_a = 10^{17}$ cm^{-3} and $N_d = 5 \times 10^{15}$ cm^{-3}. $T = 300$ K. (*a*) Calculate the Fermi level on each side of the junction with respect to the intrinsic Fermi level. (*b*) Sketch the equilibrium energy-band diagram for the junction and determine V_{bi} from the diagram and the results of part (*a*). (*c*) Calculate V_{bi} using Equation (7.10), and compare the results to part (*b*). (*d*) Determine x_n, x_p, and the peak electric field for this junction.

(Use n_i = 1.5 ×10^{10} cm^{-3} and ε = 11.7 for silicon at 300K)

[0.3294 eV, 0.407 eV, 0.7364 V, 0.7363 V, 0.426 μm, 0.0213 μm, 3.29 × 10^4 V/cm]

3.

(*a*) Consider a uniformly doped silicon pn junction at $T = 300$ K. At zero bias, 25 percent of the total space charge region is in the n-region. The built-in potential barrier is $V_{bi} = 0.710$ V. Determine (*i*) N_a, (*ii*) N_d, (*iii*) x_n, (*iv*) x_p, and (*v*) $|E_{max}|$. (*b*) Repeat part (*a*) for a GaAs pn junction with $V_{bi} = 1.180$ V.

(Use n_i = 1.5 ×10^{10} cm^{-3} and ε = 11.7 for silicon and n_i = 1.8 ×10^6 cm^{-3} and ε = 13.1 for GaAs)

[7.766 × 10^{15} cm^{-3}, 2.33 × 10^{16} cm^{-3}, 0.0993 μm, 0.2979 μm, 3.58 × 10^4 V/cm,

8.127×10^{15} cm^{-3}, 2.438×10^{16} cm^{-3}, 0.1324 μm, 0.3973 μm, 4.45×10^4 V/cm]

4.

(*a*) The peak electric field in a reverse-biased silicon pn junction is $|E_{max}| = 3 \times 10^5$ V/cm. The doping concentrations are $N_d = 4 \times 10^{15}$ cm^{-3} and $N_a = 4 \times 10^{17}$ cm^{-3}. Find the magnitude of the reverse-biased voltage. (*b*) Repeat part (*a*) for $N_d = 4 \times 10^{16}$ cm^{-3} and $N_a = 4 \times 10^{17}$ cm^{-3}. (*c*) Repeat part (*a*) for $N_d = N_a = 4 \times 10^{17}$ cm^{-3}.

(Use $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\varepsilon = 11.7$ for silicon)

[0.766 V, 72.8 V, 0.826 V, 7.18 V, 0.886 V, 0.57V]

5.

*A Si p-n junction has donor and acceptor doping concentrations N_d and N_a of 10^{16} cm^{-3} and 10^{18} cm^{-3}, respectively. Assume kT = 0.026 eV at 300 K, $n_i = 1.5\times10^{10}$ cm^{-3} and $\varepsilon_r = 11.8$. Calculate under thermal equilibrium conditions

a) the Fermi energy levels and draw an equilibrium energy band diagram.
b) the contact potential V_o (compare it to that obtained in part (a))
c) the depletion width W
d) the penetration depth into the n-side and p-side
e) peak electric field ξ_o at the junction and
f) the space charge in each side of the depletion region, given a cross-sectional area of 10^{-4} cm^2

Comment on depletion width and the penetration depth obtained in relation to the doping concentration on each side. **(Ans: a) $E_{ip} - E_{Fp} = 0.468$ eV, $E_{Fn} - E_{in} = 0.349$ eV, b) 0.817 V, c) 0.328 μm, d) $x_{po} = 3.25\times10^{-3}$ μm, $x_{no} = 0.325$ μm, e) -5×10^4 V/cm, f) 5.2×10^{-12} C)**

6.

Calculate the forward bias V_F of an ideal p-n junction diode at 300 K to result in a current of 15 mA if the reverse saturation current I_o is (a) 5 μA and (b) 8 pA. **(0.2 V, 0.53 V)**

7.

*A Si p-n junction has donor and acceptor doping concentrations N_d and N_a of 10^{16} cm^{-3} and 10^{18} cm^{-3}, respectively. Assume kT = 0.026 eV at 300 K, $n_i = 1.5\times10^{10}$ cm^{-3} and $\varepsilon_r = 11.8$. Calculate the following parameters at (i) FB of 0.65 V and (ii) RB of 2 V.

a) contact potentials under both FB and RB. Draw the band diagrams. Clearly indicate the extent of band bending.
b) the depletion width W
c) the penetration depth into the n-side and p-side
d) peak electric field ξ_o at the junction and
e) the space charge in each side of the depletion region, given a cross-sectional area of 10^{-4} cm^2

(Ans: a) V_c = 0.167 V, V_c = 2.817 V, b) 0.148 μm, 0.610 μm, c) 1.47×10^{-3} μm, 0.147 μm, 6.04 $\times10^{-3}$ μm, 0.604 μm, d) -2.251×10^4 V/cm, -9.25×10^4 V/cm, e) 2.35 pC, 9.66 pC)

8.

Consider a silicon p-n junction with donor and acceptor doping concentrations N_d and N_a of 10^{15} cm^{-3} and 10^{18} cm^{-3}, respectively. A forward bias of 0.52 V is applied to the p-n junction. Calculate the minority-carrier concentrations at the depletion layer-edges. Assuming a very weak electric field in the neutral regions of the semiconductor (outside the depletion region), estimate the ratio of the majority carrier to minority carrier drift currents at the depletion layer edge and deep inside the n-region. Assume that the diode is at room temperature and μ_n = 1350 cm^2/V-s, μ_p = 450 cm^2/V-s and $n_i = 1.5\times10^{10}$ cm^{-3}. **(10^{14} cm^{-3}, 10^{11} cm^{-3}, 30, 1.34×10^{10})**

9.

*An abrupt Si p-n junction diode is fabricated with 1 Ω-cm p-type and 0.2 Ω-cm n-type materials in which the minority carrier lifetimes are $\tau_n = 10^{-6}$ sec and $\tau_p = 10^{-8}$ sec respectively. A forward bias of 0.55 V is applied across the junction.

a) Calculate the donor concentration of the n-type material and the acceptor concentration of the p-type material
b) Calculate the minority carrier diffusion lengths
c) Calculate the density of minority carriers at both edges of the space charge region
d) Sketch the majority and minority carrier distributions as functions of distance from the junction indicating the minority carrier diffusion lengths on your distance axis

e) Sketch the majority and minority carrier currents outside the space charge region as functions of distance from the junction

f) Calculate the location of the plane at which the majority carrier and minority carrier currents are equal in magnitude

Assume that T = 300K, kT = 0.026 eV, and for Si, μ_n = 0.145 m^2/V-sec, μ_p = 0.045 m^2/V-sec and $n_i = 1.5\times10^{10} cm^{-3}$.

(Ans : a) 2.16×10^{22} m^{-3}, 1.39×10^{22} m^{-3}, b) 6.14×10^{-5} m, 3.42×10^{-6} m, c) 1.6×10^{19} m^{-3}, 2.49×10^{19} m^{-3}, f) 1.53×10^{-6} m from the edge at the n-side)

10.

A silicon p-n junction diode has the following parameters at 300 K

$D_n = 25$ cm^2/sec $\quad$ $D_p = 10$ cm^2/sec $\quad$ $\tau_n = \tau_p = 0.5$ μs $\quad$ $\varepsilon_r = 11.7$
$n_i = 1.5\times10^{10}$ /cm^3

Given that the electron diffusion current is 20 A/cm^2 and the hole diffusion current is 5 A/cm^2 at a forward bias of 0.65 V, determine the doping densities of the p and the n regions of the diode. **($N_a = 10^{15}$ /cm^3 and $N_d = 2.5\times10^{15}$ /cm^3)**